

Engineering Notes

Curvature-Constrained Lyapunov Vector Field for Standoff Target Tracking

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I. Introduction

TARGET tracking techniques allow aircraft to achieve surveillance objectives like convoy protection, area surveillance, and distributed atmospheric volume sampling. Tracking is commonly achieved by following a target while maintaining a constant standoff distance, which is known as standoff target tracking. This can be achieved using approaches based on the “good helmsman” behavior [1–3], virtual target-based path-following guidance [4–7], or relative side-bearing angle [8]. The convergence of the unmanned aerial vehicle (UAV) onto the tracking path can also be achieved through a vector field-based path-following approach. With a simple position-based guidance command, vector fields present a popular guidance framework for standoff target tracking. The vector field provides a desired heading at every point in the loiter plane to guide the UAV onto the tracking path. A good standoff target tracking technique satisfies the surveillance objective while considering the limits within which the aircraft can fly. Specifically, the aircraft should maintain a minimum standoff distance from the target so as to avoid its detection. More important, while reaching the standoff trajectory, the UAV turn rate should be within the capability imposed by the bank angle limits. Designing guidance laws by considering the aircraft constraints leads to a realistic surveillance capability.

Lawrence [9] introduced the Lyapunov vector field-based guidance law for standoff target tracking. The vector field was derived using a Lyapunov function that satisfied the minimum energy condition at the tracking path and an energy penalty elsewhere in the loiter plane. Lawrence et al. [10,11] and Jankovic et al. [12] presented the mathematical framework for deriving the families of such vector fields. This vector field provided the desired heading at a point in the loiter plane based on the Lyapunov function. The vector field, at each point, is a resultant of the contraction term that governs the radial speed and the circulation term that controls the tangential speed with respect to the target. This work was extended by Frew and Lawrence [13] to have multiple UAVs track a moving target with a Lyapunov vector field-based approach together with a speed control for achieving phasing between the aircrafts. Experimental validation of the results of [13] was discussed in [14], and a control structure for

cooperative standoff tracking of a moving target was presented in [15]. Note that, by a Lyapunov-based construction, Lyapunov vector fields are globally asymptotically stable.

Nelson et al. [16,17] introduced another approach to derive vector fields wherein the tangent of the desired heading at a location was proportional to the positional error with respect to the desired path. Griffiths [18] built upon the work of [17] and obtained the vector field for an arbitrary curved path. Note that, unlike Lyapunov vector fields, the stability is not inherently guaranteed for these vector fields and Lyapunov’s direct approach was used to prove the stability. Chen et al. [19,20] presented the tangent plus Lyapunov vector field, which used a tangent vector field on the outside of the standoff circle and a Lyapunov vector field inside. Lim et al. [21] presented a numerical integration method for calculating individual UAV paths to achieve simultaneous capture. This was carried out by using a guidance parameter calculated based on the initial and terminal goal positions.

Note that, inside the standoff circle, the curvature for the conventional Lyapunov vector field is maximum at the center [9,11,13–15]. For surveillance in self-territory, the UAVs often take off near the center of the standoff circle and will ideally desire a zero curvature near it. This work presents a new Lyapunov vector field based on a circulation parameter that is a function of the radial distance from the target. The resulting curvature characteristics are analyzed and closed-form relations are obtained for the maximum curvature. The curvature approaches zero both near the center of the standoff circle and a distance far away from it. With respect to existing Lyapunov vector fields, two major advantages are presented. First, with a lower maximum curvature limit, the proposed method provides flyable trajectories both inside and outside the standoff circle. Second, a faster convergence is established using the contraction term analysis. Furthermore, simulations are carried out for tracking a stationary target and a moving target in the presence of wind from both inside and outside the standoff circle.

The remainder of the Note is arranged as follows: Sec. II describes the standoff tracking problem. The proposed vector field construction is presented in Sec. III. Simulation studies are discussed in Sec. IV. Section V contains concluding remarks.

II. Problem Statement

Consider a UAV tracking a stationary target as shown in Fig. 1. The UAV has a position vector $\mathbf{r} = [x \ y]^T$ in the fixed frame with the target at $[0 \ 0]^T$. The location of the UAV can also be represented using polar coordinates with r and θ as the magnitude and direction of $\mathbf{r}$, respectively. A normal to the loiter plane (or x - y plane) is represented by $\hat{\mathbf{n}}$. Flying in the loiter plane, the UAV has a velocity $\mathbf{v}$ with a course χ . The UAV is to be guided onto the standoff circle of radius r_d by controlling the course to track a desired guidance vector $\hat{\mathbf{r}}$. Prospective guidance vectors are desired to provide a fast convergence to the standoff circle while taking into consideration the turn rate limits of the UAV.

III. Proposed Lyapunov Vector Field

A. Vector Field Construction

Lawrence et al. [11] presented a systematic way to construct a family of vector fields from the Lyapunov function. The desired attractor $\mathcal{A}$ is specified at the standoff circle as the zero-level set of an otherwise positive scalar potential function $V(\mathbf{r}): \mathbb{R}^2 \rightarrow \mathbb{R}_{\geq 0}$, where $\mathbb{R}_{\geq 0} = \{p \in \mathbb{R} | p \geq 0\}$. Hence, the function $V(\mathbf{r})$ satisfies following conditions:

- 1) $V(\mathbf{r})$ is positive semidefinite on all $\mathbf{r}$ with $V(\mathbf{r}) = 0$ when $\mathbf{r} \in \mathcal{A}$.
- 2) $V(\mathbf{r})$ is radially unbounded.

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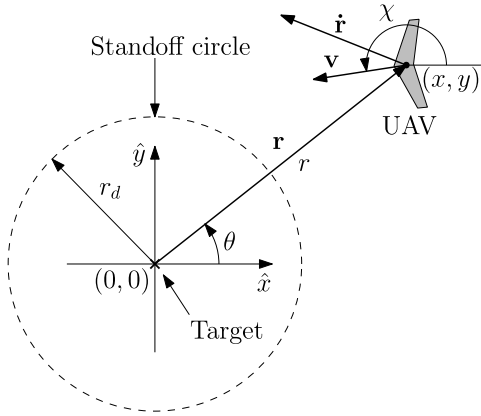


Fig. 1 Standoff tracking scenario.

3) $V(\mathbf{r})$ is continuously differentiable for all $\mathbf{r} \in \mathbb{R}^2$ and has no local extrema. Thus,

$$\frac{dV(\mathbf{r})}{d\mathbf{r}} = \mathbf{0} \Rightarrow \mathbf{r} \in \mathcal{A}$$

4) $V(\mathbf{r})$ is not an explicit function of time.

A function $V(\mathbf{r})$ that satisfies the preceding conditions is chosen as

$$V(\mathbf{r}) = V(x, y) = (x^2 + y^2 - r_d^2)^2 \quad (1)$$

Note that the attractor

$$\mathcal{A} = \{[x \ y]^T \in \mathbb{R}^2 | x^2 + y^2 = r_d^2\} = \{\mathbf{r} \in \mathbb{R}^2 | r = r_d\}$$

is at the standoff circle, where $V(x, y) = 0$. On differentiating $V(\mathbf{r})$ with respect to time,

$$\dot{V}(\mathbf{r}) = \frac{dV(\mathbf{r})}{d\mathbf{r}} \cdot \dot{\mathbf{r}} \quad (2)$$

where a change in $V(\mathbf{r})$ with respect to $\mathbf{r}$ is given by

$$\begin{aligned} \frac{dV(\mathbf{r})}{d\mathbf{r}} &= 4(x^2 + y^2 - r_d^2)[x \ y] \\ &= 4(r^2 - r_d^2)\mathbf{r}^T \end{aligned} \quad (3)$$

which is obtained by substituting for the position vector $\mathbf{r}$ that satisfies $x^2 + y^2 = r^2$. Moreover, the vector $\dot{\mathbf{r}}$ from Eq. (2) is the desired Lyapunov guidance vector defined as

$$\dot{\mathbf{r}} = -\left[\frac{dV(\mathbf{r})}{d\mathbf{r}}\Gamma(\mathbf{r})\right]^T + \mathbf{S}(\mathbf{r}) \quad (4)$$

Here, the term

$$-\left[\frac{dV(\mathbf{r})}{d\mathbf{r}}\Gamma(\mathbf{r})\right]^T$$

is called the contraction component, denoted by $\mathbf{A}(\mathbf{r})$, where $\Gamma(\mathbf{r}): \mathbb{R}^2 \rightarrow \mathbb{R}^{2 \times 2}$ is a scaled identity matrix defined as

$$\Gamma(\mathbf{r}) = \frac{1}{\alpha(\mathbf{r})}\mathbb{I}_{2 \times 2}$$

with $\alpha(\mathbf{r})$ as the speed normalization term and $\mathbb{I}$ as the 2×2 identity matrix. Hence,

$$\begin{aligned} \mathbf{A}(\mathbf{r}) &= -\left[\frac{dV(\mathbf{r})}{d\mathbf{r}}\Gamma(\mathbf{r})\right]^T = -\frac{4(r^2 - r_d^2)}{\alpha(\mathbf{r})}(\mathbf{r}^T\mathbb{I})^T = -\frac{4(r^2 - r_d^2)}{\alpha(\mathbf{r})}\mathbf{r} \\ &= -\frac{4r(r^2 - r_d^2)}{\alpha(\mathbf{r})}\hat{\mathbf{r}} \end{aligned} \quad (5)$$

where $\hat{\mathbf{r}}$ is the unit vector along $\mathbf{r}$. Furthermore, $\mathbf{S}(\mathbf{r}): \mathbb{R}^2 \rightarrow \mathbb{R}^2$, called the circulation component, is such that

$$\frac{dV(\mathbf{r})}{d\mathbf{r}} \cdot \mathbf{S}(\mathbf{r}) = 0 \quad (6)$$

Clearly, $\mathbf{S}(\mathbf{r})$ is always normal to

$$\frac{dV(\mathbf{r})}{d\mathbf{r}}$$

and it does not affect change in $V(\mathbf{r})$ over time. On comparing with Eq. (3), $\mathbf{S}(\mathbf{r})$ is also perpendicular to $\mathbf{r}$ as seen from Fig. 2. As an important condition, $\mathbf{S}(\mathbf{r}) \neq \mathbf{0}$ at $\mathbf{r} \in \mathcal{A}$. Hence, $\mathbf{S}(\mathbf{r})$, which is tangential with respect to origin in the loiter plane, is defined as

$$\mathbf{S}(\mathbf{r}) = \frac{\gamma}{\alpha(\mathbf{r})}\hat{\mathbf{n}} \times \hat{\mathbf{r}} \quad (7)$$

where $\gamma > 0$. Figure 2 shows the contraction and circulation components for the UAV at a location $\mathbf{r}$, the resultant of which provides the desired guidance vector field $\dot{\mathbf{r}}$. Subsequently, γ in Eq. (7) is chosen as

$$\gamma = 4c r_d r^2 \quad (8)$$

where $c > 0$. For a UAV flying at constant speed s , using Eqs. (5), (7), and (8), the speed normalization term is given by solving for $\alpha(\mathbf{r})$ in the constrained equation

$$\sqrt{\|\mathbf{A}(\mathbf{r})\|^2 + \|\mathbf{S}(\mathbf{r})\|^2} = s \quad (9)$$

Hence, $\alpha(\mathbf{r})$ is obtained as

$$\begin{aligned} \alpha(\mathbf{r}) &= \frac{4r\sqrt{(r - r_d)^2 + (crr_d)^2}}{s} \\ &= \frac{4r\sqrt{r^4 + (c^2 - 2)r_d^2r^2 + r_d^4}}{s} \end{aligned} \quad (10)$$

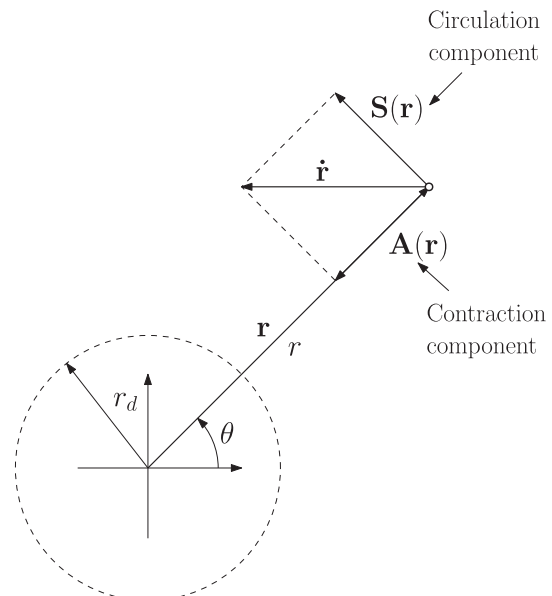


Fig. 2 Contraction and circulation components for a UAV at $\mathbf{r}$.

Using Eqs. (5), (7), (8), and (10) in Eq. (4), the desired Lyapunov vector field is given by

$$\dot{\mathbf{r}} = -\frac{s(r^2 - r_d^2)}{\sqrt{r^4 + (c^2 - 2)r_d^2 r^2 + r_d^4}} \hat{\mathbf{r}} + \frac{scr_d r}{\sqrt{r^4 + (c^2 - 2)r_d^2 r^2 + r_d^4}} \hat{\mathbf{n}} \times \hat{\mathbf{r}} \quad (11)$$

In Eq. (11), the contraction term governs the rate of change of the radial distance r with respect to time as given by

$$\dot{r} = -\frac{s(r^2 - r_d^2)}{\sqrt{r^4 + (c^2 - 2)r_d^2 r^2 + r_d^4}} \quad (12)$$

Similarly, the value of the circulation term directs the UAV in the tangential direction as governed by

$$r\dot{\theta} = \frac{scr_d r}{\sqrt{r^4 + (c^2 - 2)r_d^2 r^2 + r_d^4}} \quad (13)$$

The radial and tangential unit velocity vectors in the loiter plane can be expressed in the Cartesian frame as

$$\hat{\mathbf{r}} = \cos \theta \hat{\mathbf{x}} + \sin \theta \hat{\mathbf{y}} \quad (14)$$

$$\hat{\mathbf{n}} \times \hat{\mathbf{r}} = -\sin \theta \hat{\mathbf{x}} + \cos \theta \hat{\mathbf{y}} \quad (15)$$

Substituting $\cos \theta = (x/r)$ and $\sin \theta = (y/r)$ in Eqs. (14) and (15) and using Eq. (11), the Lyapunov vector field in Cartesian coordinates takes the form

$$\dot{\mathbf{r}} = -\frac{s}{\sqrt{r^4 + (c^2 - 2)r_d^2 r^2 + r_d^4}} \left[\left(\frac{(r^2 - r_d^2)}{r} x + cr_d y \right) \hat{\mathbf{x}} + \left(\frac{(r^2 - r_d^2)}{r} y - cr_d x \right) \hat{\mathbf{y}} \right] \quad (16)$$

B. Proposed Circulation Parameter

This work considers the circulation constant c of Eq. (8) as a guidance parameter and proposes a new circulation term. The motivation here is to have a circulation term that is maximum at the standoff circle and reduces to zero as we approach either the center of the circle or a faraway point outside the circle. Hence, a potential choice of c should satisfy 1) $c(r)$ is continuous in r , 2) $c(r)$ is maximum at the standoff distance r_d for maximum circulation, and 3) $c(r) = 0$ at $r = 0$ and $c(r) \rightarrow 0$ as $r \rightarrow \infty$.

A choice of c that satisfies the aforementioned conditions is chosen as

$$c(r) = \begin{cases} \frac{r}{r_d} & \text{for } r < r_d \\ \frac{r_d}{r} & \text{for } r \geq r_d \end{cases} \quad (17)$$

Figure 3 shows a plot of $c(r)$ versus r for a standoff distance of $r_d = 10$. Figure 4a presents the Lyapunov vector field of [9], which is obtained by using $c = 2$ and $r_d = 10$ in Eq. (16). For the same r_d and the proposed $c(r)$ of Eq. (17), the resulting vector field is presented in Fig. 4b. Note that, for the proposed vector field, the vectors move radially outward from the center. A detailed analysis of the curvature characteristics of both the vector fields are carried out in Secs. III.D and III.E.

C. Contraction Term Analysis

A relatively higher value of the contraction term guarantees faster settling onto the standoff circle. The magnitude of the contraction term is obtained from Eq. (12) as

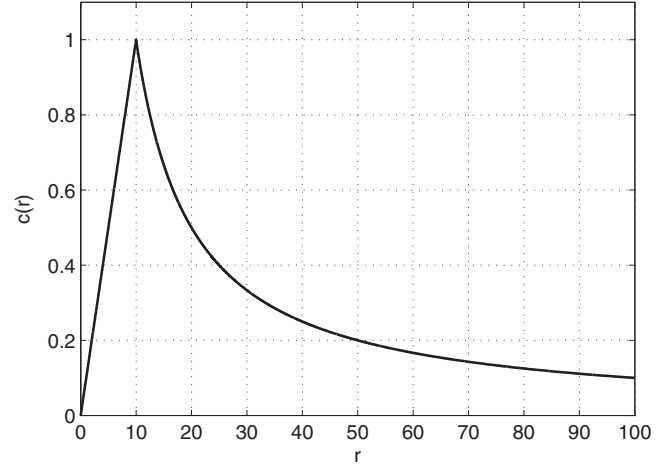


Fig. 3 Variation of proposed $c(r)$ with r for $r_d = 10$.

$$\Lambda = \frac{s|r^2 - r_d^2|}{\sqrt{r^4 + (c^2 - 2)r_d^2 r^2 + r_d^4}} \quad (18)$$

Note that the contraction term is a function of s , c , r , and r_d . To analyze the contraction characteristics as a function of r/r_d , Eq. (18) is rewritten as

$$\Lambda = \frac{s|(r/r_d)^2 - 1|}{\sqrt{(r/r_d)^4 + (c^2 - 2)(r/r_d)^2 + 1}} \quad (19)$$

For the vector field of [9], the guidance parameter $c = 2$. Substituting this value of c in Eq. (19) leads to

$$\Lambda_{c=2} = \frac{s|(r/r_d)^2 - 1|}{(r/r_d)^2 + 1} \quad (20)$$

where $\Lambda_{c=2}$ is the contraction term magnitude for the vector field of [9]. In the case of the proposed vector field, using Eq. (17) in Eq. (19), for a UAV converging from outside of the standoff circle, the guidance parameter $c = r_d/r$. Substituting this value of c in Eq. (19), the magnitude of contraction term outside the standoff circle $\Lambda_{\text{prop,out}}$ is obtained as

$$\Lambda_{\text{prop,out}} = \frac{s|(r/r_d)^2 - 1|}{\sqrt{(r/r_d)^4 - 2(r/r_d)^2 + 2}} \quad \text{for } r \geq r_d \quad (21)$$

While inside the circle, the contraction term magnitude $\Lambda_{\text{prop,in}}$ for the proposed vector field can be obtained by substituting $c = r/r_d$ from Eq. (17) in Eq. (19). The resulting expression is

$$\Lambda_{\text{prop,in}} = \frac{s|(r/r_d)^2 - 1|}{\sqrt{2(r/r_d)^4 - 2(r/r_d)^2 + 1}} \quad \text{for } r < r_d \quad (22)$$

For a comparison, the ratio of the contraction terms ξ is considered. For outside the standoff circle, the ratio ξ_{out} can be obtained using Eqs. (20) and (21) as

$$\xi_{\text{out}} = \frac{\Lambda_{\text{prop,out}}}{\Lambda_{c=2}} = \frac{(r/r_d)^2 + 1}{\sqrt{(r/r_d)^4 - 2(r/r_d)^2 + 2}} \quad \text{for } r \geq r_d \quad (23)$$

and similarly for inside the circle, Eqs. (20) and (22) give the ratio ξ_{in} as

$$\xi_{\text{in}} = \frac{\Lambda_{\text{prop,out}}}{\Lambda_{c=2}} = \frac{(r/r_d)^2 + 1}{\sqrt{2(r/r_d)^4 - 2(r/r_d)^2 + 1}} \quad \text{for } r < r_d \quad (24)$$

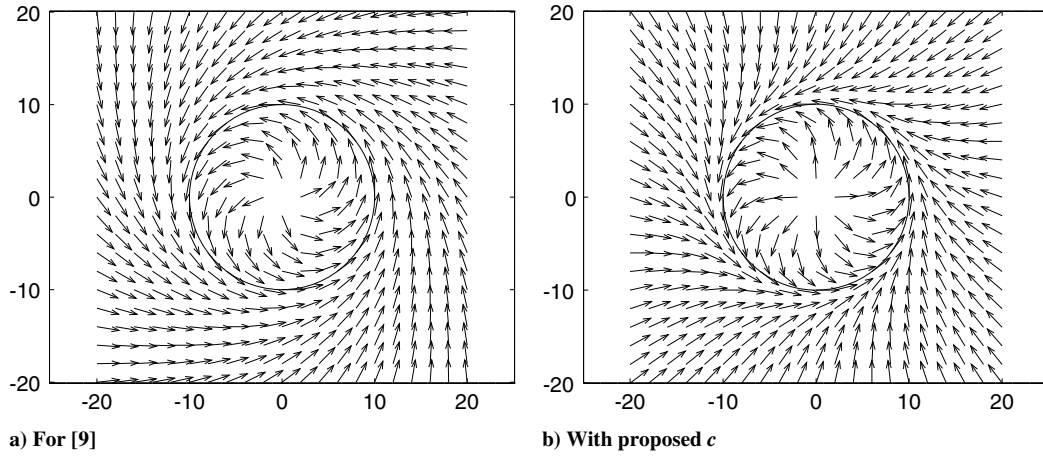


Fig. 4 Lyapunov vector field with standoff distance of $r_d = 10$.

Figure 5a presents the variation for the contraction ratio with $r \geq r_d$. Note that, near the standoff radius, the contraction ratio has a value greater than two and the ratio asymptotically converges to one with increasing r/r_d . More importantly, the contraction ratio is always greater than one for a UAV approaching from outside the standoff circle. For inside the standoff circle, the contraction ratio is plotted in Fig. 5b. At the center of the standoff circle, the contraction terms are identical for both the vector fields. While moving closer to the standoff circle, the contraction ratio increases, reaches a maximum, and then settles to two. Note that, in both the cases, the contraction ratio is always greater than or equal to one. The effect of a relatively higher contraction term on the settling is revisited in Sec. IV, wherein a much faster convergence to the standoff circle is achieved due to a higher contraction component.

D. Curvature Characteristics

This subsection analyzes in detail the curvature characteristics of the proposed vector field. The variation is analyzed and a closed-form relation for the maximum curvature is obtained. The characteristics are compared with [9].

Considering a stationary target, the course can be expressed as

$$\chi = \tan^{-1} \left(\frac{\dot{y}}{\dot{x}} \right) \quad (25)$$

which leads to the course rate $\dot{\chi}$:

$$\dot{\chi} = \frac{\dot{x}\ddot{y} - \dot{y}\ddot{x}}{\dot{x}^2 + \dot{y}^2} \quad (26)$$

The curvature κ and course rate are related through the speed of the UAV s as

$$\kappa = \frac{\dot{\chi}}{s} \quad (27)$$

Substituting Eq. (26) in Eq. (27),

$$\kappa = \frac{1}{s} \left(\frac{\dot{x}\ddot{y} - \dot{y}\ddot{x}}{\dot{x}^2 + \dot{y}^2} \right) \quad (28)$$

The x and y components of the vector field relative to the stationary frame are obtained from Eq. (16) as

$$\dot{x} = -\frac{s}{\sqrt{r^4 + (c^2 - 2)r_d^2 r^2 + r_d^4}} \left(\frac{(r^2 - r_d^2)}{r} x + c r_d y \right) \quad (29)$$

$$\dot{y} = -\frac{s}{\sqrt{r^4 + (c^2 - 2)r_d^2 r^2 + r_d^4}} \left(\frac{(r^2 - r_d^2)}{r} y - c r_d x \right) \quad (30)$$

Substituting Eqs. (29) and (30) and their time derivatives in Eq. (28) leads to

$$\kappa = \frac{s c r_d^3 ((c^2 - 2)r^2 + 2r_d^2) - \dot{c} r_d r (r^2 - r_d^2) \sqrt{r^4 + (c^2 - 2)r_d^2 r^2 + r_d^4}}{s (r^4 + (c^2 - 2)r_d^2 r^2 + r_d^4)^{3/2}} \quad (31)$$

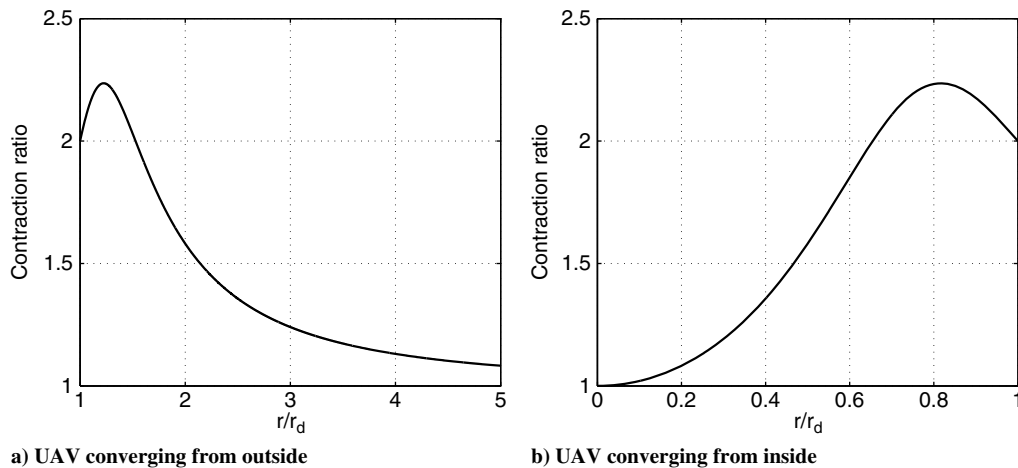


Fig. 5 Contraction term ratio of the proposed vector field to that of [9].

Using Eq. (17), the derivative $\dot{c}(r)$ is obtained as

$$\dot{c}(r) = \begin{cases} -\frac{s(r^2-r_d^2)}{r_d\sqrt{r^4+(c^2-2)r_d^2r^2+r_d^4}} & \text{for } r < r_d \\ \frac{sr_d(r^2-r_d^2)}{r^2\sqrt{r^4+(c^2-2)r_d^2r^2+r_d^4}} & \text{for } r \geq r_d \end{cases} \quad (32)$$

Using Eqs. (17), (31), and (32), the curvature for the proposed vector field is deduced as

$$\kappa_{\text{prop}} = \begin{cases} \frac{2r^5-4r_d^2r^3+3r_d^4r}{(2r^4-2r_d^2r^2+r_d^4)^{3/2}} & \text{for } r < r_d \\ \frac{r_d^2(2r_d^4-r^4)}{r(r^4-2r_d^2r^2+2r_d^4)^{3/2}} & \text{for } r \geq r_d \end{cases} \quad (33)$$

In contrast, a constant c (as used in existing Lyapunov vector fields) reduces Eq. (31) to

$$\kappa = \frac{cr_d^3((c^2-2)r^2+2r_d^2)}{(r^4+(c^2-2)r_d^2r^2+r_d^4)^{3/2}} \quad (34)$$

In particular, for the vector field of [9] with $c = 2$, the curvature $\kappa_{c=2}$ can be obtained from Eq. (34) as

$$\kappa_{c=2} = \frac{4r_d^3}{(r^2+r_d^2)^2} \quad (35)$$

Note that, using Eq. (35), the maximum curvature $\kappa_{c=2,\max}$ for the Lyapunov vector field of [9] is at $r = 0$ with

$$\kappa_{c=2,\max} = \frac{4}{r_d} \quad (36)$$

Figure 6 presents the curvature variation for the proposed Lyapunov vector field and the Lyapunov vector field of [9]. A subsequent analysis deduces the stationary points for the curvature profiles of the proposed field, finally leading to its maximum value. Differentiating κ_{prop} of Eq. (33) with respect to r ,

$$\frac{d\kappa_{\text{prop}}}{dr} = \begin{cases} \frac{-4r^8+16r_d^2r^6-20r_d^4r^4+3r_d^8}{(2r^4-2r_d^2r^2+r_d^4)^{5/2}} & \text{for } r < r_d \\ \frac{3r_d^2r^8-20r_d^6r^4+16r_d^8r^2-4r_d^{10}}{r^2(r^4-2r_d^2r^2+2r_d^4)^{5/2}} & \text{for } r \geq r_d \end{cases} \quad (37)$$

Equating

$$\frac{d\kappa_{\text{prop}}}{dr}$$

to zero, two sets of equations are obtained for $r < r_d$ and $r \geq r_d$.

For $r < r_d$,

$$-4r^8 + 16r_d^2r^6 - 20r_d^4r^4 + 3r_d^8 = 0 \quad (38)$$

$$\Rightarrow -4\left(\frac{r}{r_d}\right)^8 + 16\left(\frac{r}{r_d}\right)^6 - 20\left(\frac{r}{r_d}\right)^4 + 3 = 0 \quad (39)$$

Solving Eq. (39), the value of (r/r_d) for the extremum of κ is obtained as

$$\frac{r}{r_d} = 0.6891 \quad (40)$$

For $r \geq r_d$,

$$3r_d^2r^8 - 20r_d^6r^4 + 16r_d^8r^2 - 4r_d^{10} = 0 \quad (41)$$

$$\Rightarrow 3\left(\frac{r}{r_d}\right)^8 - 20\left(\frac{r}{r_d}\right)^4 + 16\left(\frac{r}{r_d}\right)^2 - 4 = 0 \quad (42)$$

Subsequently, by solving Eq. (42), the other extremum of κ is achieved at

$$\frac{r}{r_d} = 1.4512 \quad (43)$$

Substituting for (r/r_d) of Eq. (43) in Eq. (33), we obtain

$$\kappa_{\text{extremum}} = \begin{cases} \frac{3.0126}{r_d} & \text{for } r = 0.689r_d \\ -\frac{0.5062}{r_d} & \text{for } r = 1.451r_d \end{cases} \quad (44)$$

Equation (44) shows that the magnitude of the curvature extremum is higher inside the standoff circle. Hence, for a given standoff radius, the overall maximum curvature for the proposed vector field is

$$\kappa_{\text{prop,max}} = \frac{3.0126}{r_d} \quad (45)$$

Considering all arbitrary positions, the corresponding maximum turn rate requirement for the UAV can be obtained using Eqs. (27) and (45) as

$$\dot{\chi}_{\max} = s\kappa_{\max} \quad (46)$$

$$= \frac{3.0126s}{r_d} \quad (47)$$

E. Curvature Comparison

Comparing Eqs. (36) and (45), a major advantage is achieved in terms of the maximum curvature. The proposed vector field presents a significantly lower maximum curvature ($\kappa_{\text{prop,max}} = (3.0126/r_d)$) as compared to [9] ($\kappa_{c=2,\max} = (4/r_d)$). Furthermore, the curvature gradually increases from zero [see Eq. (33) and Fig. 6] for trajectories starting near the center of the standoff circle. This presents a major advantage in terms of generating flyable trajectories starting inside the standoff circle. In contrast, the curvature is maximum [see Eqs. (35) and (36), and Fig. 6] at the center for [9]. For all the trajectories starting from outside the standoff circle, using Eq. (33) for $r \geq r_d$ and Eq. (35), both the vectors fields have the maximum curvature demand of $(1/r_d)$ at the standoff circle.

F. Further Comparisons with Varying Choice of c

The conventional vector field of [9] considers $c = 2$. In this subsection, a comparison is carried out by considering different choices for the constant circulation parameter c .

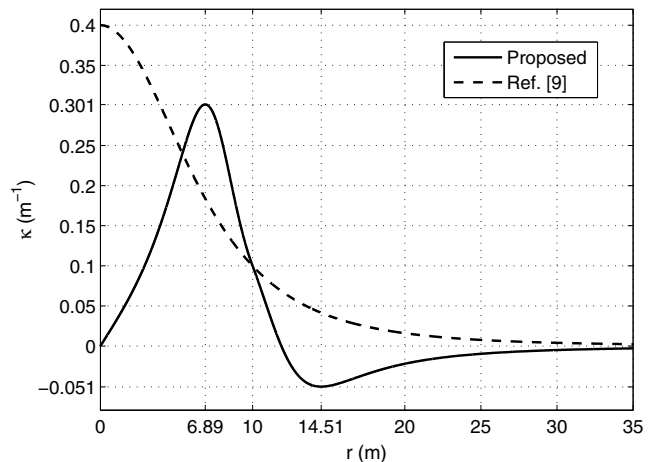


Fig. 6 Vector field curvature comparison for $r_d = 10$ m.

1. With c Greater Than or Equal to One

Figure 7 shows the comparison results with $c \geq 1$ for a convergence from outside the standoff circle with $r_d = 10$ m. Figure 7a presents the contraction ratio variation. Note that, for all the cases with $c \geq 1$, the average contraction ratio greater than one that corresponds to a faster convergence for the proposed method. Furthermore, an increased c leads to a higher contraction ratio, and hence an increased advantage in terms of faster convergence. The corresponding curvature profiles are shown in Fig. 7b. For all the vector fields under consideration, the maximum curvature occurs at the standoff circle. For constant c curvature profiles, the variation is smoother for higher values of c . Next, a comparison is carried out for positions inside the standoff circle for $c \geq 1$. The contraction ratio comparison is shown in Fig. 8a where a similar advantage is achieved in terms of a faster settling. Figure 8b presents the corresponding curvature comparison. As a unique feature, the proposed vector field provides a zero curvature at the center of the standoff circle that cannot be obtained by a constant c .

2. With $0 < c < 1$

A similar comparison is carried out by considering $c \in (0, 1)$. The resulting contraction ratio variation is presented in Fig. 9. For smaller c , the average contraction ratio can reduce to less than one, and hence provide a faster settling. However, as seen from Fig. 10, with a smaller choice of c , the curvature demand increases rapidly. Note that, in between, there could be some choices of c that provide a faster settling with a lower maximum curvature demand. However, as seen from Fig. 10b, this would require a considerable curvature at the center of the standoff circle. The detailed comparison highlights a

major advantage of the proposed vector field construction in terms of simultaneously satisfying maximum curvature constraints and zero curvature at the standoff center.

G. Stability

The stability of the proposed vector field is verified using the Lyapunov stability criteria. From Eq. (1), the Lyapunov function is

$$V(\mathbf{r}) = V(x, y) = (x^2 + y^2 - r_d^2)^2$$

Differentiating $V(\mathbf{r})$ with respect to time,

$$\dot{V}(\mathbf{r}) = \frac{dV(\mathbf{r})}{d\mathbf{r}} \cdot \dot{\mathbf{r}} \quad (48)$$

Substituting for the Lyapunov vector field $\dot{\mathbf{r}}$ from Eq. (4) into Eq. (48) and using Eq. (6),

$$\begin{aligned} \dot{V}(\mathbf{r}) &= \frac{dV(\mathbf{r})}{d\mathbf{r}} \cdot \left(-\left[\frac{dV(\mathbf{r})}{d\mathbf{r}} \Gamma(\mathbf{r}) \right]^T + S(\mathbf{r}) \right) \\ &= -\left(\frac{dV(\mathbf{r})}{d\mathbf{r}} \right) \Gamma(\mathbf{r}) \left(\frac{dV(\mathbf{r})}{d\mathbf{r}} \right)^T + \frac{dV(\mathbf{r})}{d\mathbf{r}} \cdot S(\mathbf{r}) \\ &= -\left(\frac{dV(\mathbf{r})}{d\mathbf{r}} \right) \Gamma(\mathbf{r}) \left(\frac{dV(\mathbf{r})}{d\mathbf{r}} \right)^T \end{aligned} \quad (49)$$

Because $\Gamma(\mathbf{r})$ is positive definite, $\dot{V}(\mathbf{r}) \leq 0$ for all $\mathbf{r} \in \mathbb{R}^2$ with $\dot{V}(\mathbf{r}) = 0$ when $\mathbf{r} \in \mathcal{A}$. From Eq. (49), using Lyapunov's direct

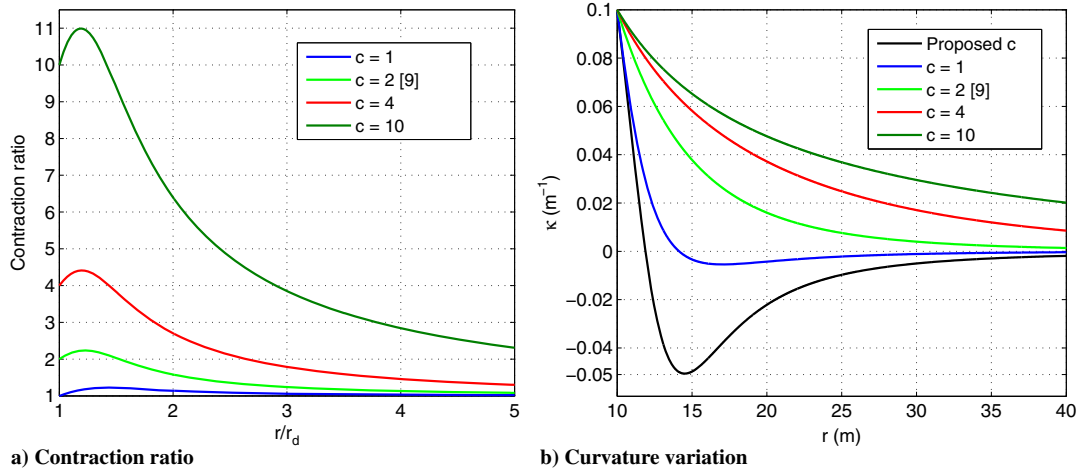


Fig. 7 Comparisons with a constant $c \geq 1$; convergence from outside, $r_d = 10$ m.

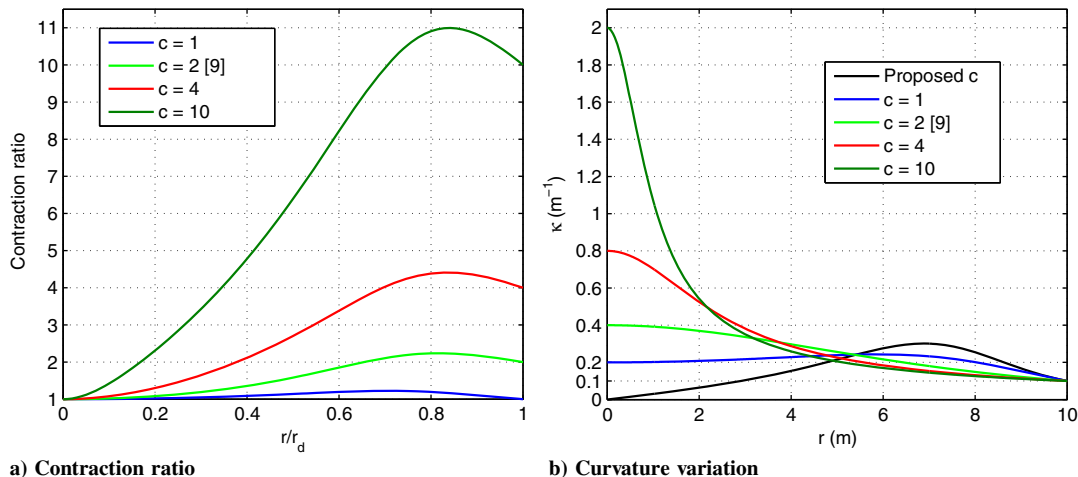
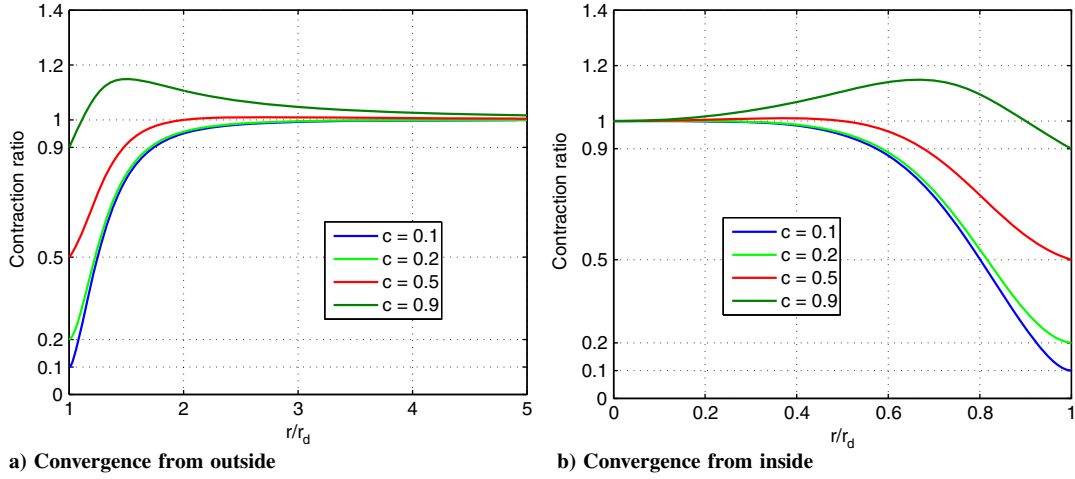
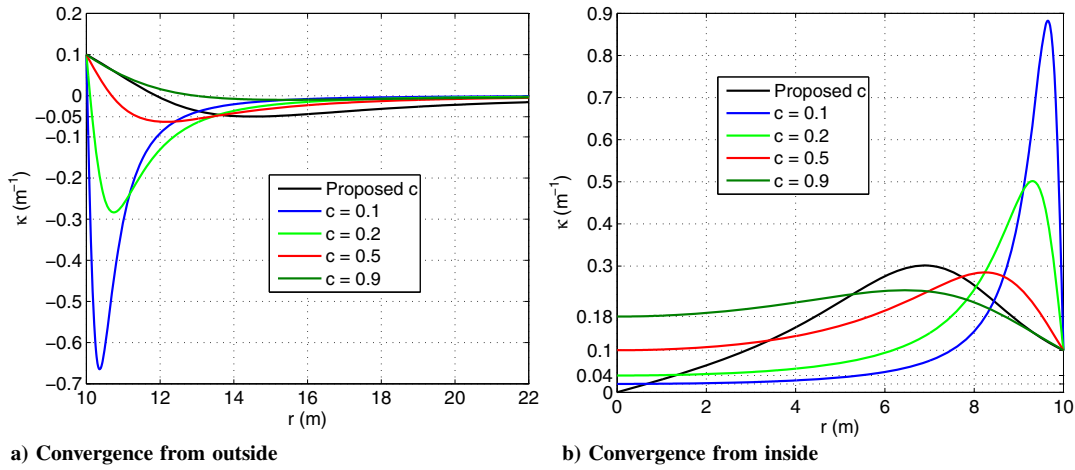


Fig. 8 Comparisons with a constant $c \geq 1$; convergence from inside, $r_d = 10$ m.

Fig. 9 Contraction ratio for a constant $c \in (0,1)$ Fig. 10 Curvature variation for a constant $c \in (0,1)$ with $r_d = 10$ m.

method, it can be concluded that the system is asymptotically stable about the equilibrium set $\mathcal{A}$.

IV. Simulations

Simulations are carried out by assuming a first-order course-hold loop for the autopilot given by

$$\dot{\chi} = \alpha(\chi_c - \chi) \quad (50)$$

where α is the control gain, and χ_c is the commanded course. Considering a coordinated turn, the corresponding bank angle demand can be expressed as

$$\tan \phi = \frac{s}{g} \dot{\chi} \quad (51)$$

The guidance laws based on the proposed Lyapunov vector field and the vector field of [9] are simulated using a constant-speed UAV model with a speed of $s = 10$ m/s and $\alpha = 200$. The maximum bank angle limit for the aircraft is taken as $\phi_{\max} = \pm 45$ deg. Using Eqs. (47) and (51), the corresponding maximum turn rate and curvature limits are $\dot{\chi}_{\max} = 0.981$ rad/s and $\kappa_{\max} = 0.098$ m^{-1} , respectively.

A. Case 1: Stationary Target with UAV Initial Position Inside the Standoff Circle

As governed by Eqs. (33), (37), and (44), while inside the standoff circle, the curvature increases from zero near the center to a maximum

of $\kappa = (3.0126/r_d)$ at $(r/r_d) = 0.689$ and decreases to $\kappa = (1/r_d)$ at $r = r_d$. A flyable path can be generated if the maximum curvature is within the UAV capability. A standoff radius can be decided based on this constraint using Eq. (45) as

$$r_d \geq \frac{3.0126}{\kappa_{\max}} = 30.71 \text{ m}$$

Here, the limiting case is chosen with $r_d = 30.71$ m. The UAV's initial position is close to the center of the standoff circle. Figure 11a shows the UAV trajectory generated by the proposed vector field. The UAV converges to the standoff distance $r_d = 30.71$ m, as shown in Fig. 11b. As shown in Fig. 11c, the curvature initially increases, reaches the maximum, and then decreases as the UAV moves toward the standoff circle. The maximum curvature of $\kappa = 0.098$ m^{-1} is attained at $r = 21.16$ m. Note that, at this point, $(r/r_d) = 0.689$, which complies with the analytic result of Eq. (44). The corresponding bank angle profile is shown in Fig. 11d, wherein the maximum bank of $\phi = 45$ deg is applied at $(r/r_d) = 0.689$. Figure 11e presents the heading angle characteristics with a close match for commanded and achieved profiles with first-order course dynamics. For the same initial conditions and parameters, a comparison is carried out with the Lyapunov vector field guidance law of [9]. The comparative UAV trajectory is shown in Fig. 11f. Figure 11g compares the standoff distance variation for both the vector fields. Note that, owing to a higher contraction term as analyzed in Sec. III.C, the proposed vector field settles faster onto the standoff circle. Corresponding curvature variations are plotted in Fig. 11h. The curvature profile for [9] has the maximum at the center

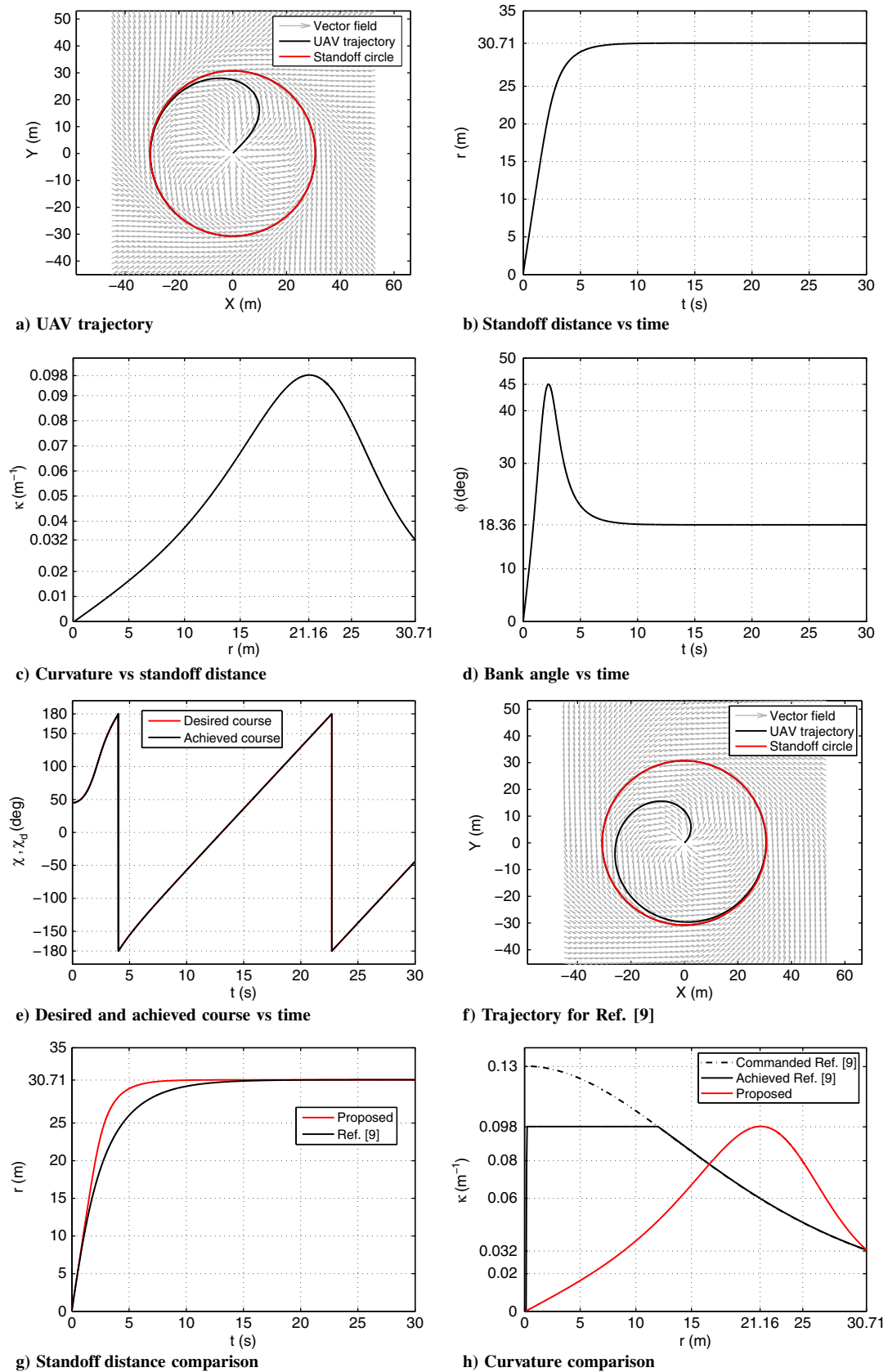


Fig. 11 Results for case 1: stationary target with UAV initial position inside the standoff circle.

and it reduces with an increasing r . However, due to turning rate limitations, the UAV is not able to follow the commanded curvature and saturates at the maximum achievable value. With the commanded curvature reducing below the saturation value, the UAV then follows

it closely. On the other hand, the proposed method generates flyable UAV trajectories well within the turn rate limits. And second, as the contraction component is relatively much larger away from the standoff circle, the proposed method results in a faster settling.

B. Case 2: Stationary Target with UAV Initial Position Outside the Standoff Circle

From the analysis presented in Secs. III.D and III.E, for a UAV starting from outside the standoff circle, the curvature of the proposed vector field reduces and attains a local minimum at $(r/r_d) = 1.4512$ with $\kappa = -(0.5062/r_d)$ and then increases to $\kappa = (1/r_d)$ at the standoff circle. Note that the absolute maximum curvature in this case occurs at the standoff circle. Hence, for a flyable trajectory, the desired standoff radius should satisfy

$$r_d \geq \frac{1}{\kappa_{\max}} = 10.19 \text{ m} \quad (52)$$

The simulation is carried out by choosing the limiting case with $r_d = 10.19 \text{ m}$. The UAV trajectory for the proposed vector field is plotted in Fig. 12a, wherein the proposed vector field guides the UAV to the circle. A comparative trajectory for the vector field of [9] is presented in Fig. 12b. Note that, as shown in Fig. 12c, the proposed vector field leads to a faster convergence to the standoff circle. This is again due to the higher contraction ratio for $r \geq r_d$ as

discussed in Sec. III.C. Figure 12d shows the curvature for the vector field of [9] that monotonically increases with decreasing r until the standoff distance r_d . Note that, in this case, both the fields require an identical maximum curvature of $1/r_d$. However, a faster settling time is achieved by the proposed vector field due to a higher range rate as shown in Fig. 12e. The stationary point of $\kappa = -(0.5062/r_d) \text{ m}^{-1}$ at $(r/r_d) = 1.4512$ from Fig. 12e matches the analytical result of Eq. (44). Figure 12f shows the bank angle variation wherein the maximum of $\phi_{\max} = 45^\circ$ is used at the standoff circle.

C. Case 3: Tracking Moving Targets in the Presence of Wind

Studies thus far have considered the target to be stationary. This special case considers tracking a moving target in the presence of wind. Figure 13a shows the UAV initially at $[20 \ 0]^T \text{ m}$ with the target at the origin. The desired standoff circle radius is $r_d = 30.71 \text{ m}$. The wind and target velocities are given by $[0.2 \ -1]^T \text{ m/s}$ and $[3.5 \ 2.5]^T \text{ m/s}$, respectively. The dotted circle in Fig. 13a represents the initial standoff circle. The straight line

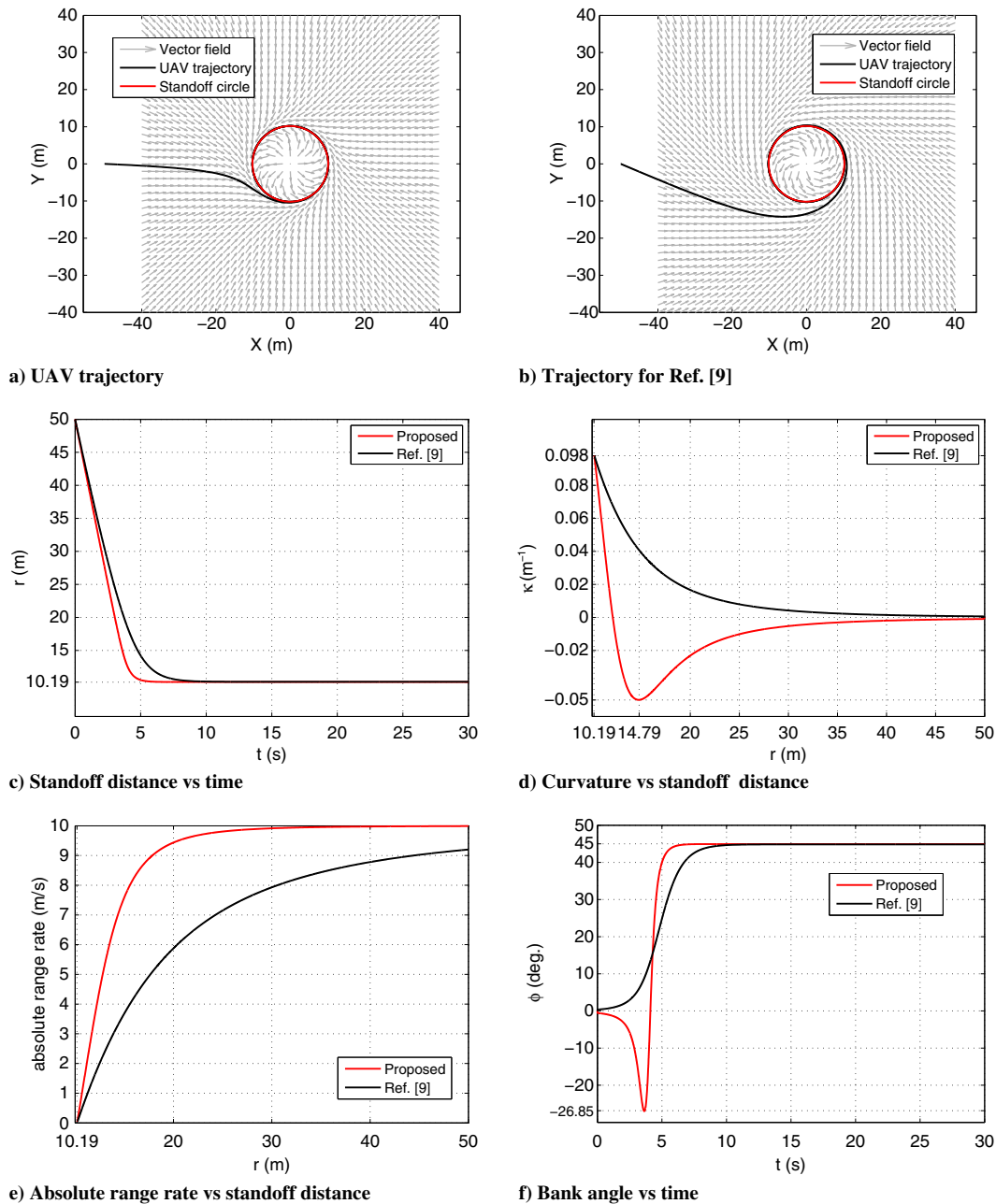


Fig. 12 Results for case 2: stationary target with UAV initial position outside the standoff circle.

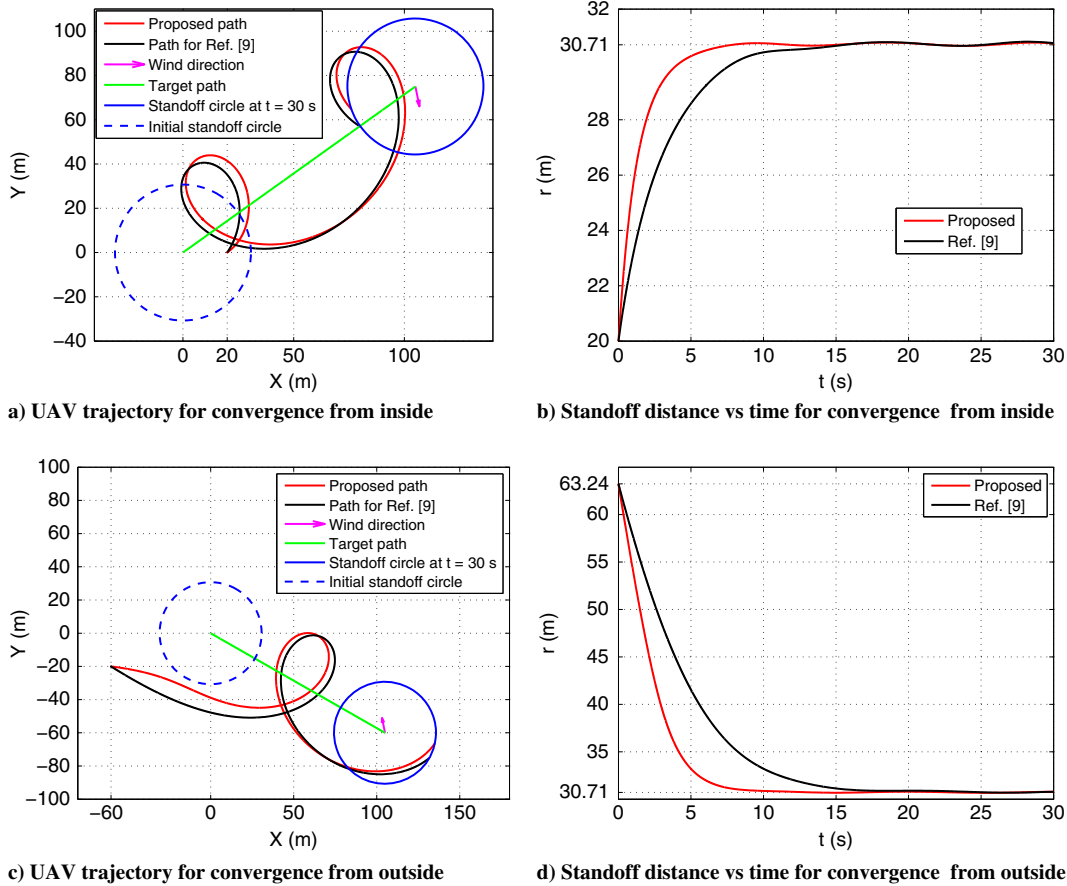


Fig. 13 Results for case 3: tracking moving targets in the presence of wind.

represents the target path over time. An instantaneous standoff circle of radius r_d is considered around the target. The path evolves over time as the vector field guides the UAV onto this instantaneous circle. From Fig. 13b, the UAV reaches and follows the standoff distance of $r_d = 30.71$ m with negligible errors. Another scenario is considered in Fig. 13c wherein the UAV tracks a moving target starting from an initial position outside the standoff circle. The aircraft is initially located at $[-60, -20]^T$ m with the target at the origin. The target velocity is $[3.5, -2]^T$ m/s, and the wind velocity is $[-0.21]^T$ m/s. The corresponding standoff distance as a function of time is plotted in Fig. 13d, presenting a successful convergence to the standoff circle. The comparative results generated using [9] are shown using solid lines in Fig. 13. In both the cases, the proposed method presents a faster settling onto the standoff circle. The results highlight the viability of the proposed vector field as applied to tracking moving targets.

V. Conclusions

A Lyapunov vector field with a modified circulation term is proposed for standoff target tracking. The vector field is directed radially, both at the center and far from the standoff circle. The resulting contraction term is analyzed, presenting a faster settling as compared to the conventional vector field. A detailed curvature analysis provides a closed-form solution for the maximum curvature for the proposed vector field. This maximum value is considerably lower as compared to the conventional method. Simulations are carried out to demonstrate tracking scenarios of the unmanned aerial vehicle in the presence of a stationary target without wind and a moving target in the presence of wind both inside and outside the standoff circle. As the major contribution, the proposed vector field provides an elegant construction that simultaneously increases the converging rate, lowers the curvature demand, and provides zero curvature at the standoff center.

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